Milestone 4

Project Title

HotelRevAI – AI Driven Revenue Analysis for Hotels

Abstract

The hospitality industry produces large volumes of operational and booking data every day. Analyzing this data effectively helps hotel managers understand revenue patterns, optimize pricing strategies,
and improve occupancy rates. This project, HotelRevAI, presents an AI-driven revenue analysis system developed
using Microsoft Power BI. The system transforms raw hotel booking data into meaningful insights through
data cleaning, modeling, DAX-based KPI calculations, and interactive dashboards.

The project integrates multiple analytical components including revenue monitoring, booking behavior analysis,
cancellation trend evaluation, and forecasting. Using Power Query for data transformation and DAX for metric
creation, several key performance indicators such as Total Revenue, Total Bookings, Occupancy Rate, Average Daily
Rate (ADR), RevPAR, and Cancellation Rate were calculated. Interactive dashboards were designed to help hotel
management visualize performance trends and make data-driven decisions.

The results demonstrate that business intelligence tools can significantly enhance revenue monitoring and
operational efficiency in the hospitality sector. The developed dashboard provides actionable insights that
support strategic planning, demand forecasting, and improved revenue management.

1. Introduction

The hospitality industry relies heavily on data-driven decision making to improve revenue and operational performance. Hotels collect large amounts of data related to bookings, cancellations, room
availability, customer types, and pricing. However, without proper analytics systems, it becomes difficult to
extract meaningful insights from this data.

HotelRevAI is designed to address this challenge by using business intelligence techniques to analyze hotel booking data. The project utilizes Microsoft Power BI to transform raw booking information into an interactive dashboard that presents revenue insights and performance indicators.

By combining data preprocessing, analytical modeling, and visualization techniques, this system enables hotel managers to monitor revenue patterns, identify booking trends, and evaluate operational efficiency. The project focuses on providing a structured analytical framework that supports improved decision-making and revenue optimization.

2. Problem Statement

Hotels often experience fluctuations in revenue due to unpredictable booking patterns, high cancellation rates, and inefficient pricing strategies. Traditional analysis methods depend on manual reporting or basic spreadsheet calculations, which are time-consuming and prone to errors.

There is a need for an intelligent analytics system that can automatically process booking data, generate key business metrics, and present them through intuitive dashboards. Such a system can help management monitor performance, identify problems early, and implement better strategies for revenue growth.

3. Objectives

- Analyze hotel revenue performance using structured data analytics.
- Measure booking trends and occupancy levels.
- Evaluate cancellation behavior and its impact on revenue.
- Develop key performance indicators using DAX.
- Design interactive dashboards for data visualization.
- Support data-driven decision making in hotel management.

4. Tools and Technologies

- Microsoft Power BI Desktop
- Power Query for data transformation
- DAX (Data Analysis Expressions) for KPI calculation
- Excel dataset for hotel booking data
- Data modeling techniques for analytical relationships

5. Dataset Description

The dataset used in this project contains structured booking information collected from hotel reservation systems. The dataset includes attributes such as booking ID, hotel name, city, room type,
customer category, check-in and check-out dates, booking status, revenue details, and payment methods.

These attributes enable multidimensional analysis across different business aspects including geography, time
periods, customer segments, and room categories.

6. Methodology

The project methodology consists of several analytical stages:

Data Collection – The hotel booking dataset was imported into Power BI.

Data Cleaning – Power Query was used to remove duplicates, handle missing values, and standardize data formats.

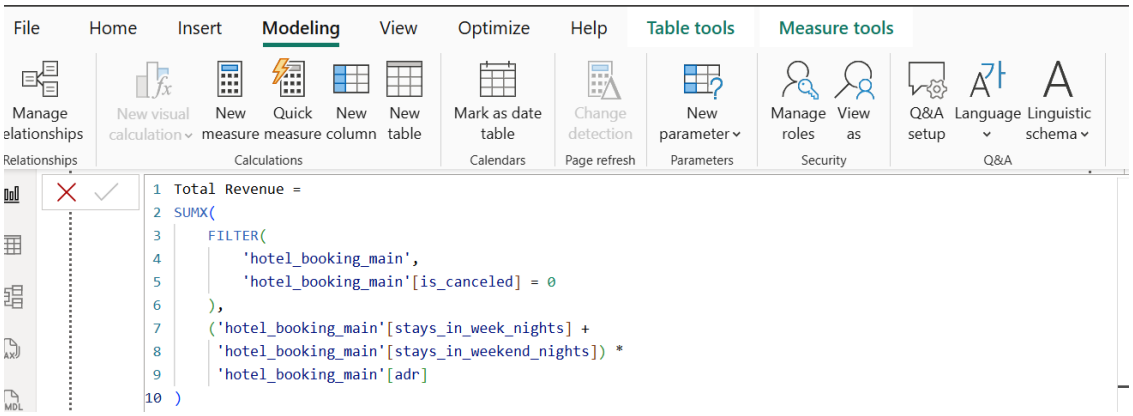
Data Modeling – Relationships between tables were established using a star schema model and a date dimension.

KPI Development – Key metrics were created using DAX formulas to measure revenue, bookings, cancellations, and
other operational indicators.

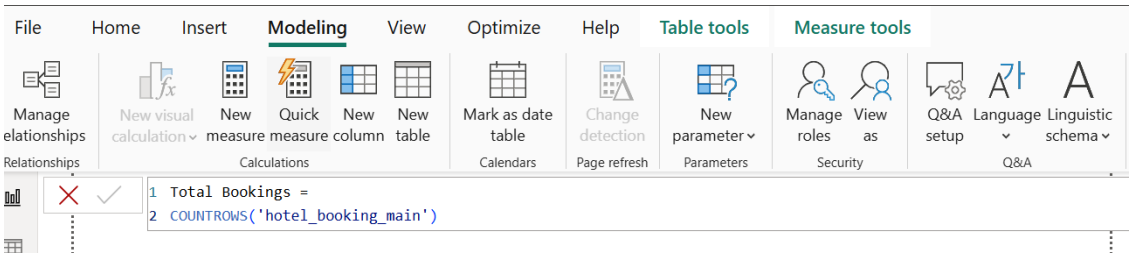
DAX Measures Implemented : -

- Total Revenue
- Total Bookings
- Total Canceled Bookings

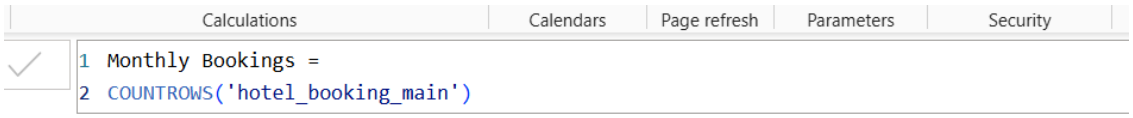
1) Total Revenue



2) Total Bookings



3) Monthly Bookings



7. Key Performance Indicators

Several important KPIs were developed to evaluate hotel performance:

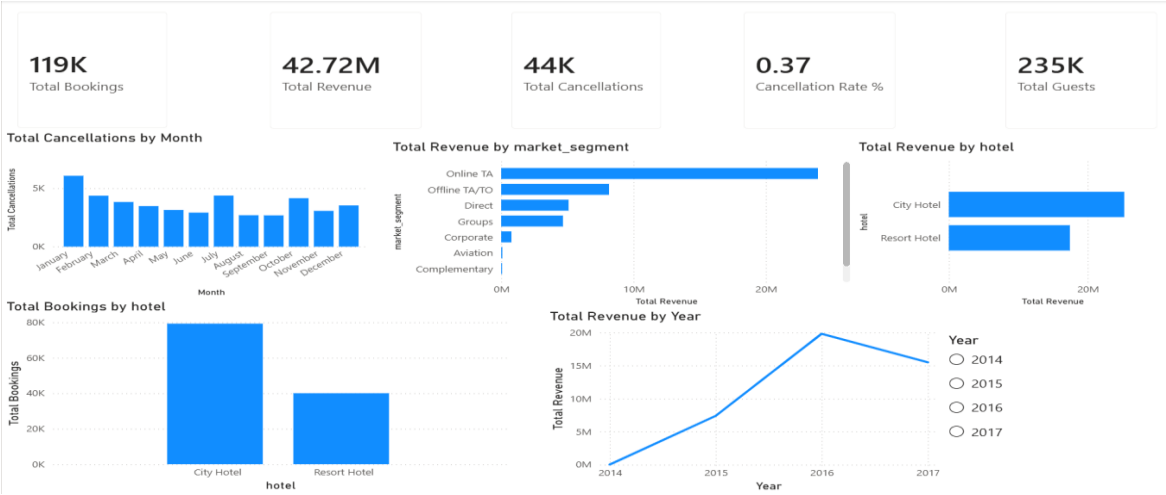
- Total Revenue – Total income generated from bookings.
- Total Bookings – Number of reservations made.
- Total Cancellations – Number of cancelled bookings.
- Cancellation Rate – Percentage of bookings cancelled.
- Average Lead Time – Average time between booking and arrival.

• Occupancy Rate – Proportion of occupied rooms.
• Average Daily Rate (ADR) – Average revenue per occupied room.
• RevPAR – Revenue generated per available room.

8. Dashboard Design

Multiple dashboard pages were created to provide comprehensive analysis.

Executive Overview – Displays KPI cards and overall revenue trends.
Revenue Analysis – Shows revenue distribution by city, hotel, and customer type.
Occupancy Analysis – Examines room utilization and booking patterns.
Cancellation Insights – Highlights cancellation trends and their financial impact.
Forecasting Dashboard – Provides future revenue projections based on historical data.



9. Business Insights

The analysis revealed several valuable insights:

• Certain cities consistently generate higher revenue.
• Premium room categories produce higher average daily rates.
• Seasonal demand significantly affects occupancy levels.
• High cancellation segments negatively impact revenue forecasting.

These insights allow hotel managers to refine pricing strategies and improve operational planning.

10. Business Impact

The HotelRevAI dashboard helps hotel management make informed decisions by providing clear and actionable analytics. The system improves revenue monitoring, identifies inefficiencies, and supports
better forecasting.

By leveraging data analytics, hotels can enhance customer satisfaction, optimize room pricing, and reduce
revenue loss caused by cancellations.

11. Future Enhancements

Future improvements to the system may include:

- Integration with real-time booking platforms
- Machine learning-based demand prediction
- Customer segmentation analysis
- Dynamic pricing models
- Predictive cancellation detection

12. Conclusion

This project successfully demonstrates how business intelligence tools can transform hotel booking data into actionable insights. Through data preprocessing, KPI development using DAX, and interactive
dashboard visualization in Power BI, the HotelRevAI system provides a comprehensive revenue analysis solution.

The dashboard enables hotel management to monitor performance trends, understand customer behavior, and
implement data-driven strategies. Overall, the project highlights the importance of analytics in improving
revenue management and operational efficiency within the hospitality industry.